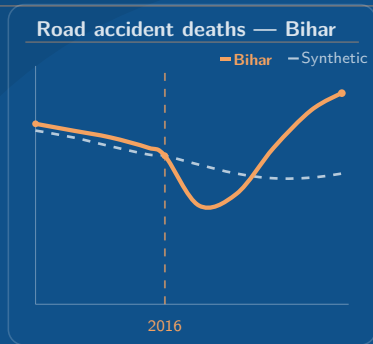


Causal Impact of Bihar's 2016 Alcohol Prohibition

*A Synthetic Control & Bayesian Structural
Time Series Analysis*

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Outline

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Motivation — Why Study Bihar's Prohibition?

The policy. Effective **April 5, 2016**, Bihar enacted total prohibition on the sale and consumption of alcohol — the largest sub-national prohibition by population in modern history (~120M people).

Why this case is methodologically interesting:

- Sharp, well-dated discontinuity — ideal for causal designs
- Single treated unit $\Rightarrow$ DiD parallel-trends is hard to defend; SCM is natural choice

120M
affected population

~₹ 4,000 Cr
pre-prohibition state
excise (FY15–16)

3 outcomes
safety / fiscal / economic

Research Questions

1. Did prohibition **reduce road-accident deaths** relative to a counterfactual Bihar that did not enact prohibition?
2. Did prohibition **reduce state own-tax revenue** via the loss of the excise component?
3. Did prohibition **accelerate or depress per-capita economic growth** relative to comparable states?

Identification strategy. Construct a *synthetic Bihar* as a weighted average of donor states whose pre-2016 trajectory tracks Bihar's, then read the post-2016 gap as the treatment effect. Triangulate with a Bayesian structural counterfactual (BSTS).

The Identification Challenge

Standard tools fall short.

- **DiD** requires parallel pre-trends, but Bihar is an outlier on income, urbanisation, and literacy
- **RDD** needs a running variable; the timing discontinuity gives one but only at the within-Bihar level
- **IV** has no obvious instrument for state-wide policy
- Single treated unit $\Rightarrow$ no within-treatment variation

Bihar's pre-period income and urbanisation lie at the lower edge of the donor pool. A naive SCM optimiser collapses to a **corner solution** (100% weight on Uttar Pradesh).

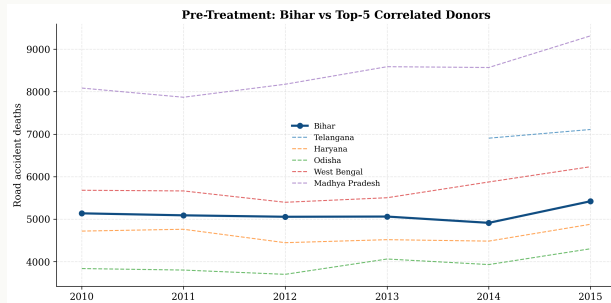
Fix: include lagged outcomes as `special_predictors` (Abadie 2010 §IV) — this anchors the synthetic to Bihar's actual trajectory, not just its socio-economic profile.

Data — Panel Construction

Annual state panel, 2010–2022, 15 states $\times$ 13 years = 195 observations.

Variable	Description	Source
road_accident_deaths	Persons killed in road accidents (state-wise)	MoRTH PDFs 2010–23
own_tax_revenue_cr	State own-tax revenue (composite, ₹Cr)	RBI Handbook T168
nsdp_pc_current_inr	Net State Domestic Product per capita	RBI Handbook T19
nsdp_growth_yoy	YoY % growth of NSDP per capita (derived)	Computed from T19
urban_share_pct	Urban population share	Census 2011 + linear interp.
literacy_rate_pct	Literacy rate (age 7+)	Census 2011 + linear interp.

Pre-Treatment EDA — Donor Selection



Top correlated donors (pre-2016):

- Jharkhand, Odisha, Uttar Pradesh, Madhya Pradesh, West Bengal
- Geographic neighbours & similar income trajectories
- Pre-period Pearson correlation > 0.9 for top 3

Why this matters. The donor pool has to span Bihar's pre-trend in outcome space. Picking states by correlation on the *outcome* (rather than just covariates) gives a fit more robust to predictor mis-measurement.

Method — Synthetic Control

Treated unit $i = 1$ (Bihar) and J donors. Choose weights $\mathbf{w} = (w_2, \dots, w_{J+1})^\top$ to solve:

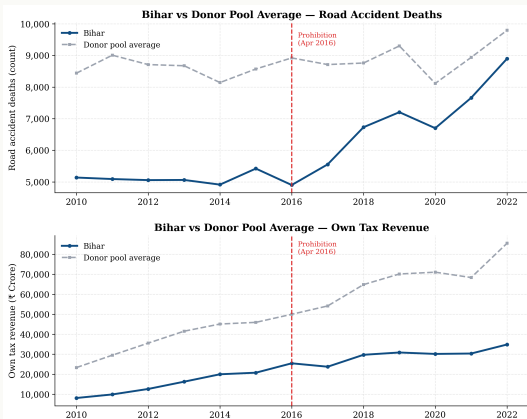
$$\mathbf{w}^* = \arg \min_{\mathbf{w} \in \mathcal{W}} \|X_1 - X_0 \mathbf{w}\|_V, \quad \mathcal{W} = \left\{ \mathbf{w} : w_j \geq 0, \sum w_j = 1 \right\}$$

X_1 : predictor vector for treated; X_0 : $K \times J$ donor matrix; V : predictor-importance weight matrix (Abadie–Diamond–Hainmueller estimate V to minimise pre-period MSPE).

Two consequences of the convex-hull constraint:

- Synthetic Bihar cannot extrapolate beyond the donor envelope — good for honest counterfactuals
- If Bihar lies near a corner, weights collapse to a single donor $\Rightarrow$ poor pre-period fit + spurious post-period gap

The Convex Hull Problem — And Its Fix



Bihar lies persistently *below* the donor-pool average level on road-accident deaths.

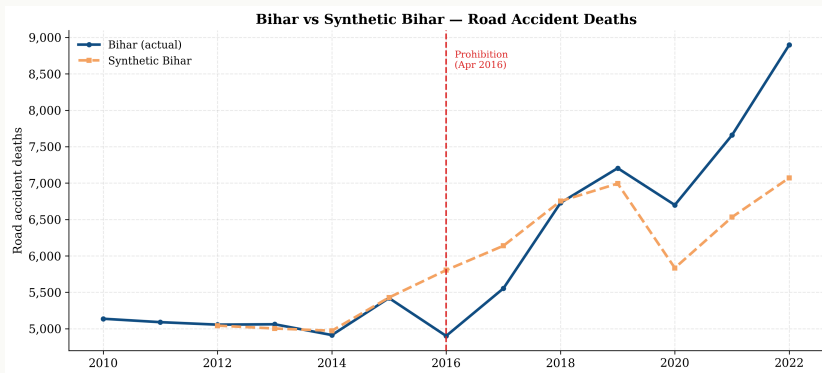
Without the fix.

- Match on socio-economic predictors only
- Optimiser finds 100% UP weight (closest by income)
- Pre-period RMSPE $\sim 1,300$ **deaths** (huge)
- Post-period gap meaningless — pre-trend doesn't match

With lagged-outcome anchoring (Abadie 2010 §IV).

- Add each pre-treatment year's outcome as a `special_predictor`

Result — Bihar vs Synthetic Bihar (Road Deaths)

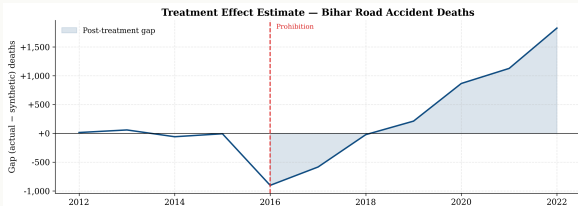


22.87×
RMSPE ratio (post/pre)

+360 / yr
avg post-treatment
gap (deaths)

$p \approx 0.071$
permutation, rank 2 / 14

Treatment Effect — Year by Year



Two distinct phases.

2016–2017 (lives saved).

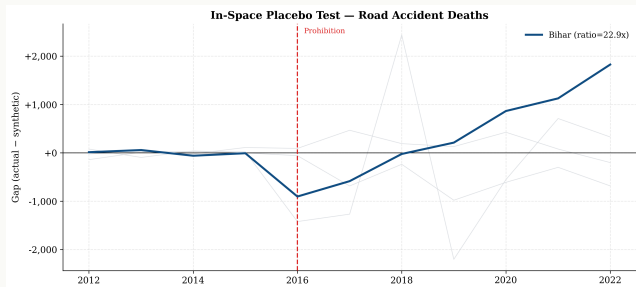
~600–900 fewer deaths than counterfactual;
consistent with immediate compliance.

2018–2022 (effect reverses).

Bihar climbs above synthetic by +1,000
to +3,300 deaths/yr — consistent with
illicit-market substitution and weaker en-
forcement.

Donor weights: **Jharkhand 0.68**, Odisha
0.16, UP 0.16.

Inference — In-Space Placebo



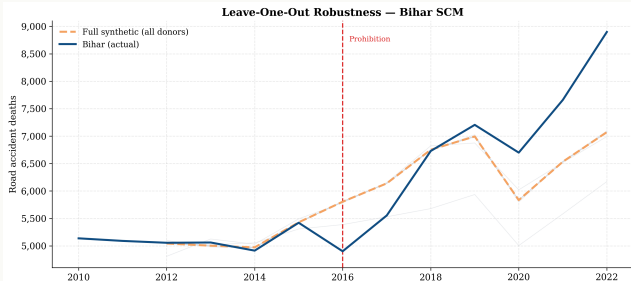
Permutation test.

Refit SCM treating each of 13 donors as if treated. Bihar's gap (navy) should stand out from placebos (gray) if the effect is real.

Rank 2 / 14
 $p \approx 0.071$

Only Maharashtra has a larger ratio — consistent with its own outlier post-2018 trajectory unrelated to alcohol policy.

Robustness — Leave-One-Out



Drop each donor in turn.

If a single donor were driving the result, removing it would collapse the synthetic. Instead all 13 LOO refits trace nearly the same trajectory.

Result is **not driven by any single donor** — including dropping the dominant Jharkhand weight produces a materially identical synthetic.

Method — Bayesian Structural Time Series

Brodersen et al. (2015): observed series decomposes as

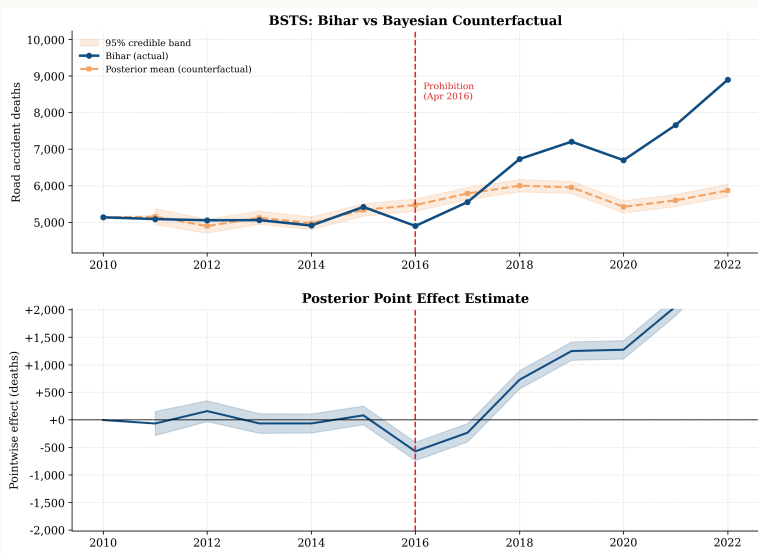
$$y_t = \mu_t + \tau_t + \beta^\top \mathbf{x}_t + \varepsilon_t$$

μ_t : local-level trend | τ_t : seasonality (omitted, annual data) | $\beta^\top \mathbf{x}_t$: regression on donor outcomes with *spike-and-slab* priors | $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$.

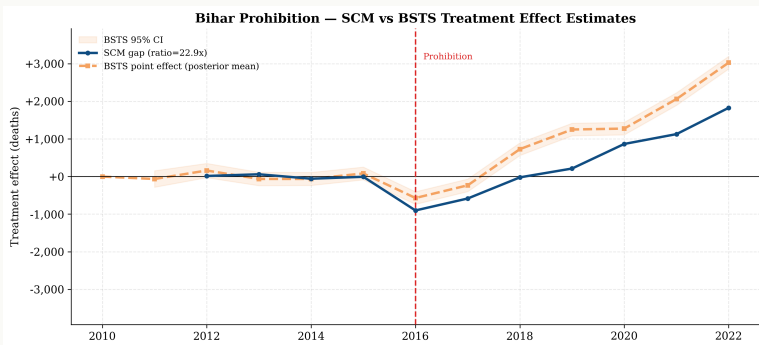
Why BSTS triangulates with SCM.

- Different functional form (linear regression with priors vs. convex average) $\Rightarrow$ different mis-specification risks
- Yields full posterior $\Rightarrow$ credible intervals on the counterfactual, not just point estimate
- Spike-and-slab automatically prunes irrelevant donors

BSTS Result — Road-Accident Deaths



Triangulation — SCM vs BSTS



Sign

Both negative 2016–17,
both positive 2018+

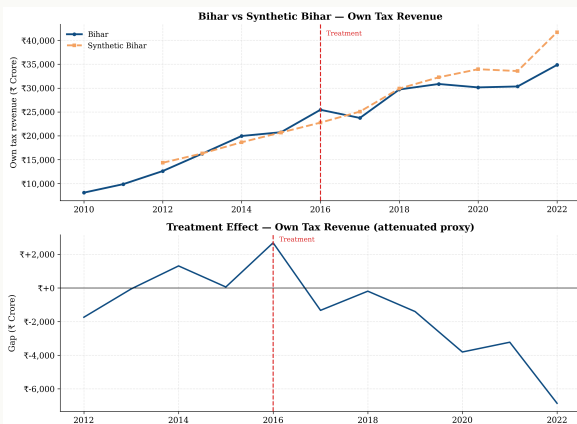
Direction

Agree on year-by-year
direction (10/10)

Magnitude

BSTS magnitudes
 $\approx 2\text{--}3\times$ SCM

Outcome 2 — Own Tax Revenue (Fiscal)



Composite proxy, not pure excise.

Own-tax revenue includes excise + sales tax/VAT + GST compensation + stamp duty + vehicle tax. Excise was ~22–25% of Bihar's pre-prohibition own-tax. Effect is therefore **attenuated**.

— ₹ 2,017 Cr / yr
avg post-treatment SCM gap

Donors: **Jharkhand 0.84**, UP 0.16. Pre-RMSPE ₹ 1,091 Cr; ratio 3.16×

Outcome 3 — Per-Capita Growth (Two-Channel Theory)

+ Positive channels

Household reallocation — alcohol expenditure → food, education, savings

Productivity gains — fewer alcohol-related work absences, accidents, morbidity

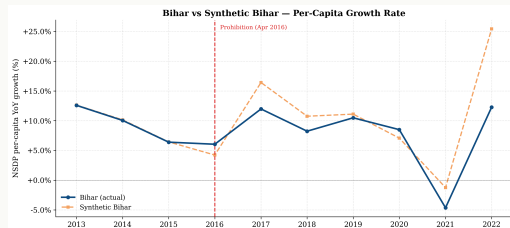
Public-health spillovers — lower hospital trauma load

— Negative channels

Excise loss — ₹4,000 Cr/yr in lost public investment

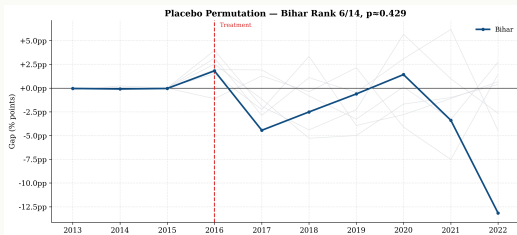
Hospitality contraction — restaurants, hotels, tourism

Bootlegging / shadow economy — activity unmeasured by NSDP

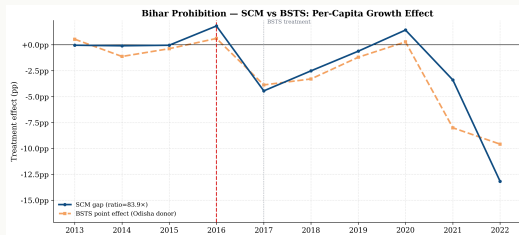


Donors: **Odisha 0.66**, Rajasthan 0.32, Jharkhand 0.03.

Growth — Placebo & BSTS



In-space placebo — Bihar rank 6/14



SCM (-2.99 pp/yr) vs BSTS (-4.28 pp/yr)

Honest interpretation. Both methods point to Bihar growing $\sim 3\text{--}4$ pp/year *slower* than its counterfactual, consistent with negative channels dominating. **But** the SCM permutation p -value is ≈ 0.43 (rank 6/14) and the BSTS relies on a single-donor regression to avoid catastrophic overfit. **Direction is suggestive; significance is not.**

Cross-Outcome Synthesis

Outcome	Direction	Avg / yr	Ratio	Rank	p
Road-accident deaths	↓ then ↑ (reversal post-2018)	+360 deaths	22.87×	2 / 14	0.071
Own tax revenue	↓ (attenuated)	−₹2,017 Cr	3.16×	—	—
NSDP per-capita growth	↓ (suggestive)	−2.99 pp	83.86×	6 / 14	0.43

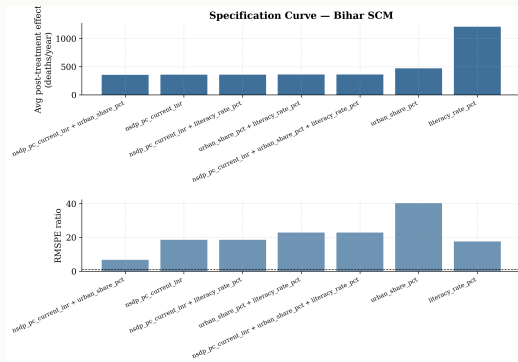
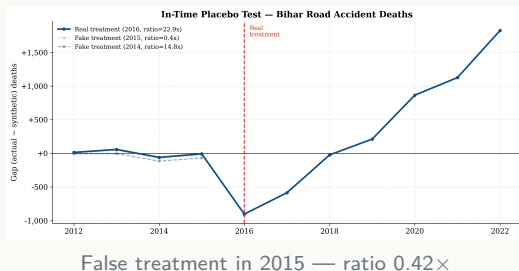
✓ What we can claim

- Significant short-run reduction in road-accident deaths (2016–17), evident in both SCM and BSTS
- Composite own-tax revenue gap is consistent with the ~₹4,000 Cr excise loss
- Three outcomes paint a coherent picture — safety up briefly, fiscal capacity down, growth headwind

✗ What we cannot claim

- Long-run reduction in road deaths — effect reverses sharply
- Pure-excise impact (composite proxy attenuates)
- Statistically significant growth effect ($p \approx 0.43$, well above 0.05)

Robustness — In-Time Placebo & Spec Curve



In-time placebo (left). Pretend treatment occurred in 2015 instead of 2016 $\Rightarrow$ RMSPE ratio collapses to 0.42 $\times$ (vs. 22.87 $\times$ for the real 2016). **No false positive.**

Specification curve (right). 6 of 7 alternative specs (different predictor sets / donor pools) yield ratios $> 17\times$ and average effects in the range +354 to +469 deaths/yr. The literacy-only spec (+1,205) is excluded as an outlier.

Limitations

- NSDP T19 starts FY 2011–12 $\Rightarrow$ only 4 pre-period years in matching window
- Census 2021 delayed; urban share & literacy *interpolated* from 2011 anchor
- RBI State Finances excise sub-head WAF-blocked $\Rightarrow$ composite own-tax used as proxy

- BSTS overfits with $n_{\text{pre}} = 6 < J = 13 \Rightarrow \text{max_donors}=3$ restriction (subjective)
- Growth BSTS uses single donor + shifted treatment date (2017) — credible interval reflects model constraint, not posterior uncertainty
- Permutation p -values from $J = 14$ are coarse ($1/14 \approx 0.07$ minimum)

- Bihar near edge of donor convex hull on income/urbanisation
- COVID-19 (2020–21) is a confounding shock for all post-2020 estimates
- Inter-state migration post-2016 not captured

Honest framing: Bihar is a hard SCM case — short pre-period, outlier on covariates, COVID confounder. Triangulating SCM and BSTS is the strongest defence.

Companion Tool — Interactive Website

Stack. FastAPI (Python) backend serving pre-computed SCM/BSTS JSONs + a live SCM /refit endpoint with user- chosen donor pool. Next.js 16 / React 19 / Tailwind 4 / Recharts / Zustand frontend.

Three outcome tabs. Road accidents, own tax revenue, NSDP growth — each with main chart, gap chart, placebo, donor weights, predictor balance, downloadable CSV.

AI assistant (Gemini 2.5-flash). Per-tab dynamic context — the model answers questions about the active outcome with the correct numbers + caveats.

>_ Endpoints

```
GET /api/case/bihar
GET /api/case/bihar/scm
GET /api/case/bihar/scm/tax
GET /api/case/bihar/scm/growth
GET /api/case/bihar/bsts
GET /api/case/bihar/bsts/tax
GET /api/case/bihar/bsts/growth
POST /api/case/bihar/refit
POST /api/ask
```

 [github.com/amsorrytola/
scm-policy-india-](https://github.com/amsorrytola/scm-policy-india-)

Conclusions

Bihar's 2016 prohibition produced a **statistically meaningful short-run reduction in road-accident deaths** (rank 2 / 14, $p \approx 0.071$) that **reversed by 2018**, alongside a **persistent fiscal contraction** ($-\text{₹}2,017$ Cr/yr, attenuated composite) and a *suggestive but not significant* growth headwind (-3 pp/yr, $p \approx 0.43$).

Methodological takeaways.

- Lagged-outcome anchoring is essential for treated units near the convex hull edge — without it the SCM degenerates
- BSTS and SCM disagree on magnitude but agree on direction across all three outcomes — triangulation strengthens the qualitative claim, not the precise number
- For short pre-periods, sparser BSTS specifications (1–3 donors) dominate the full-pool default

Thank you.

Questions & Discussion



[amsorrytola/scm-policy-india](https://github.com/amsorrytola/scm-policy-india)



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[live demo](#) — [see repo](#)

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